

Wiley Analytical Science Virtual Conference

Data integrity: Expert seminar now available on-demand

Catch-up with Dr. Bob McDowall as he will guide you through the most important aspects of instrument qualification and computer validation of analytical instruments.

Key learnings:

- Practical examples of integrated qualification and validation of analytical instruments, including chromatography data systems
- Most important steps to ensure the integrity of the data generated by these instruments and systems
- Ways of leveraging supplier development into a validation project to reduce the amount of work conducted by a laboratory

[Register here](#)



Biochemical characterization reveals the functional divergence of two tropinone reductases from *Przewalskia tangutica*

Nianyang Wu¹
 Dongqin Jian¹
 Min Xiang¹
 Min Chen²
 Xiaozhong Lan³
 Zhihua Liao ¹
 Xiaoqiang Liu ^{1*}

¹Key Laboratory of Eco-environments in Three Gorges Reservoir Region (Ministry of Education), SWU-TAAHC Medicinal Plant Joint R&D Centre, School of Life Sciences, Southwest University, Chongqing, People's Republic of China

²SWU-TAAHC Medicinal Plant Joint R&D Centre, College of Pharmaceutical Sciences, Southwest University, Chongqing, People's Republic of China

³TAAHC-SWU Medicinal Plant Joint R&D Centre, Tibetan Collaborative Innovation Centre of Agricultural and Animal Husbandry Resources, Tibet Agricultural and Animal Husbandry College, Nyingchi of Tibet, People's Republic of China

Abstract

Przewalskia tangutica is a traditional medicinal plant from Tibet used for the analgesic effect from the tropane alkaloids (TAs) produced by the plant. Its roots have the highest yield of hyoscyamine in all plant species and so have been overharvested becoming an endangered medicinal plant species. Metabolic engineering is a good way to improve the yield of TAs in plants. In our study, two functionally distinct tropinone reductases genes, *PtTRI* and *PtTRII*, were cloned from *P. tangutica* and the functional divergence were characterized. The enzyme kinetics of *PtTRI* and *PtTRII* were investigated. The phylogenetic analysis classified them into different clades: *PtTRI* and *PtTRII* were in the clade of

tropine-forming reductase and pseudotropine-forming reductase, respectively. We found *PtTRI* to be expressed in the roots but less in leaves, whereas *PtTRII* was expressed in the roots at higher levels than in the leaves. The kinetic parameters (K_m , V_{max} , and K_{cat}) were analyzed using purified recombinant enzymes at their optimum pH. Enzymatic analysis results showed that tropinone is a better substrate for *PtTRII* compared with *PtTRI*, suggesting that *PtTRII* might be a potential gene target for TA biosynthesis engineering. Compared with the reported TRIs, *PtTRI* exhibited a higher affinity for tropinone. © 2019 International Union of Biochemistry and Molecular Biology, Inc. Volume 66, Number 4, Pages 597–606, 2019

Keywords: enzyme kinetics, functional identification, *Przewalskia tangutica*, tropane alkaloids, tropinone reductase

Abbreviations: TAs, tropane alkaloids; TRI, tropine-forming reductase; TRII, pseudotropine-forming reductase; qPCR, quantitative PCR; PGK, cGMP-dependent protein kinase I; SDRs, short-chain dehydrogenases/reductases; GC–MS, gas chromatography–mass spectrometer.

*Address for correspondence: Xiaoqiang Liu, PhD, Key Laboratory of Eco-environments in Three Gorges Reservoir Region (Ministry of Education), SWU-TAAHC Medicinal Plant Joint R&D Centre, School of Life Sciences, Southwest University, Chongqing 400715, People's Republic of China. Tel. 86-23-68252365, Fax: 86-23-68252365; e-mail: liuxq@swu.edu.cn.

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Received 26 March 2019; accepted 29 April 2019

DOI: 10.1002/bab.1760

Published online 10 July 2019 in Wiley Online Library (wileyonlinelibrary.com)

1. Introduction

Przewalskia tangutica is a traditional medicinal plant species from Tibet, which is distributed in sandy and gritty grasslands of the Qinghai-Tibetan Plateau at altitudes ranging from 3,000 to 4,000 m [1]. This medicinal plant species is the original plant used to produce pharmaceutical tropane alkaloids (TAs), especially hyoscyamine. TAs are among the oldest medicinal drugs and have been used in traditional medicine for nearly 3,000 years for its prominent anticholinergic properties. These TAs are widely used in modern medicine for pain relief, anesthesia, refraining drug addiction, control of motion sickness, and for the treatment of Parkinson's disease, as anticholine drugs target the parasympathetic nervous system. Currently, the availability of TAs is entirely dependent on their isolation from plants, mainly plant species from the Solanaceae family. *P. tangutica* is being overharvested as it has very high levels

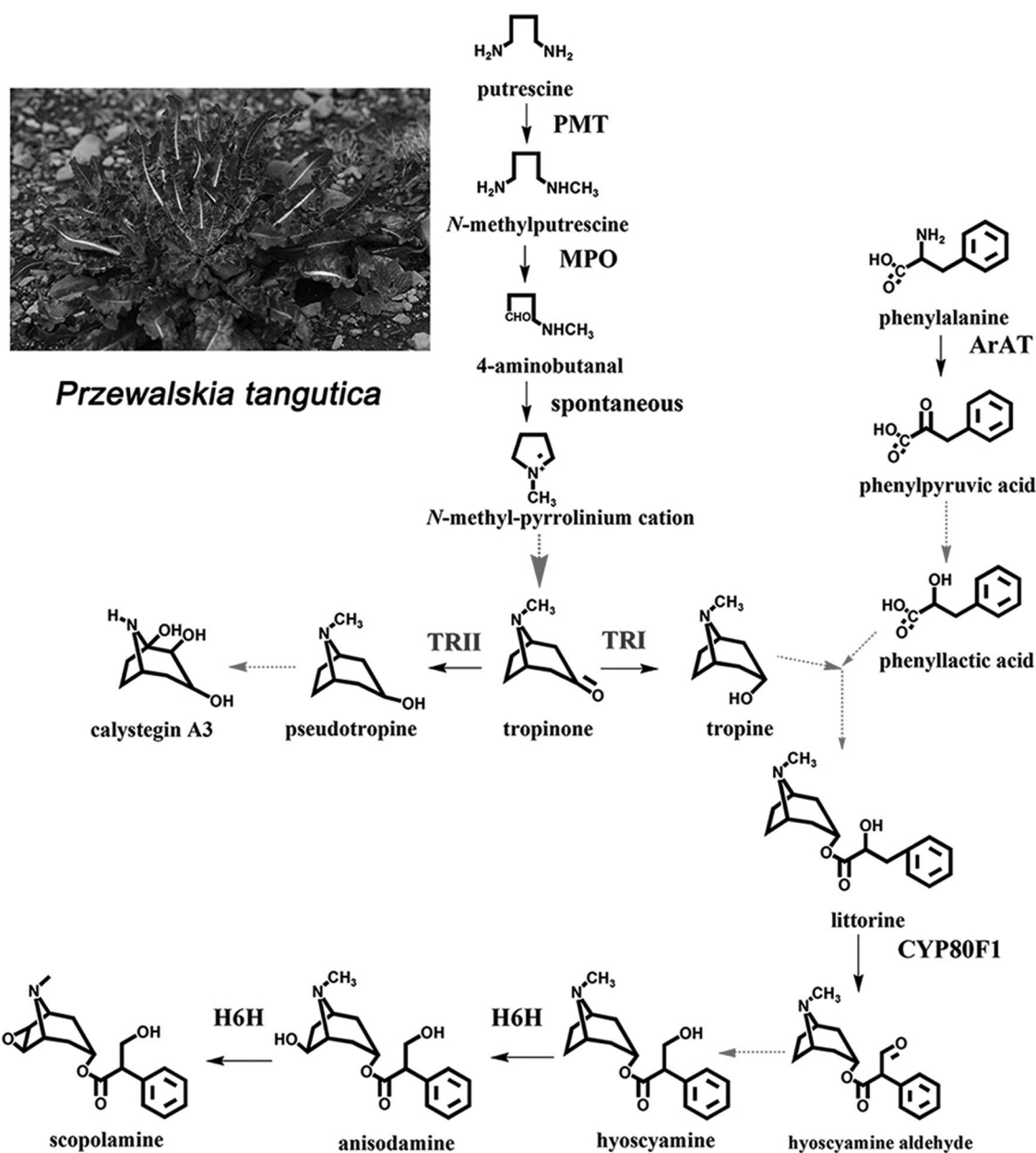


FIG. 1 The biosynthetic pathway of tropane alkaloids in the Solanaceae plant species. TRI, tropinone reductase I; TRII, tropinone reductase II.

of hyoscyamine in the roots and scopolamine in the leaves (up to 2.18% and 3.83% dry weight, respectively). Importantly, the cultivation of *P. tangutica* is still not available and currently, this valuable plant species is scarce. Among TA-producing plant species, such as *Hyoscyamus niger* [2], *Datura species* [3], *Atropa belladonna* [4], *P. tangutica* [5], and so on, *P. tangutica* gives the highest yields of TAs [1]. It is not only a valuable plant source for commercially producing hyoscyamine but also an interesting plant species for studying TA biosynthesis. Unfortunately, biosynthesis and regulation of TAs in *P. tangutica* are unknown at the molecular, biochemical, and biotechnological

level. Therefore, it is necessary to develop novel methods to increase the yield of TA using metabolic engineering [6].

Although the precise biosynthetic pathway of TAs is still unclear, several enzymes and their corresponding genes have been identified (Fig. 1). Two tropinone reductases (TRs) with a similar amino acid sequence constitute a branching point in TA metabolism. Both catalyze the stereospecific reduction of the 3-carbonyl group of tropinone to hydroxyl groups (tropine) with different stereospecific configurations [7]. Tropinone reductase I (TRI) reduces the ketone to the alcohol in the tropine ring to give products such as hyoscyamine and scopolamine, whereas pseudotropine reductase II (TRII) reduces tropinone to pseudotropine to give products of opposite configuration, such as the ones participating in the biosynthesis of nortropane alkaloids including calystegines [8]. This suggests that TRI and

TRII compete for the same substrate tropinone. TRI plays an important role in TA biosynthesis.

Generally, two different metabolic engineering approaches can promote metabolite production of the targeted substrate. One approach is overexpression of the key enzyme involved in the rate-limiting step to increase the overall flux through the biosynthetic pathway. This approach has been successfully used in increasing the yield of various pharmaceutical compounds, such as alkaloids [9, 10], terpenes [11, 12], and so on. Breaking through the pathways competitive against target metabolite biosynthesis often leads to enhanced biosynthesis of the target metabolite. Alternatively, blocking the biosynthetic pathways competing against the biosynthesis of the target metabolite is another approach but it can give unexpected results. A good example is the genetic engineering of artemisinin biosynthesis. Sterol biosynthesis is a branch pathway competing with artemisinin biosynthesis. When the first rate-limiting enzyme SQS of sterol biosynthesis was suppressed by RNAi, the sterol content significantly decreased whilst the artemisinin content dramatically increased [13]. Blocking the pseudotropine biosynthetic pathway might also facilitate the production of hyoscyamine derived from tropine [14].

In order to unveil the biosynthesis and gene regulation of TA, we identified two functionally distinct TRs, TRI and TRII, from *P. tangutica* and characterized them at molecular and biochemical levels, and compared them with the currently reported TRI, in an attempt to find a TRI with higher enzymatic activity [15]. The results showed that PtTRI might be an ideal candidate for metabolic engineering of TA alkaloids by using gene overexpression and PtTRII might be a potential gene target for engineering TA biosynthesis in *P. tangutica* through gene suppression.

2. Materials and Methods

2.1. Plant materials and treatments

The mature seeds of *P. tangutica* were harvested in the northern slope of Shergyla Mountain of Nyingchi (4,000 m, Tibet, China) in September of 2016. The seeds were germinated into pots containing a mixture of vermiculite, pindstrap moss, and perlite (6:3:1) and grown at 25 ± 1 °C under 16 h of light/8 h of darkness. When the plantlets reached about 10 cm in height, the roots, stems, and leaves of the three plants were respectively harvested for tissue profiling analysis of *PtTRI* and *PtTRII*.

2.2. Gene cloning and sequence analysis

Total RNA was extracted from the roots of *P. tangutica* using a Total RNA Kit (Tiangen, Beijing, China) according to the manufacturer's guidelines. The quality and concentration of RNA were checked using Nano spectrophotometer (Thermo Fisher Scientific, Dreieich, Germany) measurements and total RNA was subjected to reverse transcription using a cDNA Synthesis kit (Takara, Dalian, China). A pair of degenerate primers was designed based on the alignment of the TRI

nucleotide and protein sequences of the reported Solanaceae species, to carry out PCR amplification using a cDNA library of *P. tangutica* roots as a template. Based on the ensuing amplified fragments, the gene-specific primers were then used to amplify the *PtTRI*. *PtTRII* was also obtained in the same way. According to the manufacturer's protocol, gene-specific primers were designed for the rapid amplification of cDNA ends (RACE) based on the core cDNA fragment sequence information. The primer pairs for 3'-RACE PCR were: forward, PtTRI-3-1 and PtTRII-3-1 for the first 3'-end PCR amplification, PtTRI-3-2 and PtTRII-3-2 for the nested amplification; reverse, the universal primers (UPM and NUP) provided by the kit. The primer pairs for each 5'-RACE PCR reaction were as follows: forward, the universal primers (UPM and NUP) provided by the kit; reverse, PtTRI-5-1 and PtTRII-5-1 for the first 5'-end amplification, PtTRI-5-2 and PtTRII-5-1 for the nested amplification. Finally, the physical sequences of *PtTRI* and *PtTRII* were respectively amplified, using gene-specific primers. For *PtTRI*, F-*PtTRI*-full and R-*PtTRI*-full were used and for *PtTRII*, F-*PtTRII*-full, and R-*PtTRII*-full were used. Each PCR product was subcloned into the pMD-19T vector and sequenced. The primers used to isolate the full length of the genes of *PtTRI* and *PtTRII* are listed in Supplementary Table S1.

Comparative and bioinformatics analyses of *PtTRI* and *PtTRII* were carried out online at the websites <http://www.ncbi.nlm.nih.gov> and <http://www.expasy.org>. The amino acid sequence was deduced by ORFfinder and their theoretical molecular weights were calculated using Vector NTI software. BLASTP was used to analyze the sequence similarity (<http://blast.ncbi.nlm.nih.gov/Blast.cgi>). The phylogenetic tree was constructed via the neighbor-joining method using MEGA software [14, 16].

2.3. Gene expression analysis

The *P. tangutica* plants were grown in a light culture room for 2 months and the roots, stems, and leaves were isolated from the three plants with the same growth state, and the RNA of the three tissue parts was extracted. The first strand of cDNA was obtained as a template for real-time quantitative PCR (qPCR) detection, and the cGMP-dependent protein kinase I (PGK) gene was selected as the internal reference gene to calibrate the expression of *PtTRI* and *PtTRII*, according to literature methods [17]. The qPCR kits were purchased from Bio-Rad and the qPCR system was IQ5 thermal cycler (Bio-Rad, California, USA). The method of $2^{-\Delta\Delta CT}$ was used to calculate the relative gene expression levels [18]. The data were collected from three biological repeats. The primers used for gene expression analysis are listed in Supplementary Table S2.

2.4. Recombination and purification of PtTRI and PtTRII

The *PtTRI* and *PtTRII* coding region were amplified using the pair of primers PtTRI-PF, PtTRI-PR and PtTRII-PF, PtTRII-PR, respectively, and then inserted into the prokaryotic expression plasmid pET-28a between *Bam*HI and *Sac*I restriction sites, which were introduced into the host cells of *Escherichia coli*

Rosetta to express the recombinant PtTRI and PtTRII proteins. Bacteria were cultured in Luria-Bertani liquid medium with 50 mg/L kanamycin and 34 mg/L chloramphenicol at 37 °C. When the optical density (OD) value of bacterial cultures reached 0.6, the expression of PtTRI and PtTRII were induced by addition of isopropyl β -D-1-thiogalactopyranoside at a final concentration of 0.5 mM. Bacteria were further cultured at 25 °C for 7 H and then harvested for protein purification. The recombinant His-tagged PtTRI and PtTRII were purified using Ni²⁺-chelating resin columns according to the methods described in our previous publication [19]. The primers used for constructing protein recombination vector are listed in Supplementary Table S3.

2.5. Enzymatic assays

Reduction reaction system: 200 μ M NADPH, 1 mM tropinone, recombinant protein, various pH buffers up to 600 μ L (pH range 5.8–9.4). The reduction reaction was performed to detect changes in absorbance at 340 nm, and each pH measurement was replicated three times. The tropinone feeding assays were used to identify the functions of the two recombinant TRs. The purified protein of PtTRI and PtTRII were tested in 10 mL reaction volume (100 mM tropinone, 200 μ M NADPH, 100 mM buffer of potassium phosphate and 5 μ g recombinant protein) and incubated for 2 H at 30 °C. The reaction products were analyzed by gas chromatography–mass spectrometer (GC–MS) (QP2010; Ultra, Shimadzu, Kyoto, Japan). The GC–MS was performed using the same conditions with the methods previously described [4]. The kinetic constants were calculated with a nonlinear regression of the Michaelis–Menten equation using GraphPad Prism 7. All assays were repeated three times, and mean values with standard deviations are reported.

3. Results

3.1. Molecular cloning and sequence analysis

The cDNA of *PtTRI* was 1,179 bp in length that encoded a 31.2 kDa polypeptide with 274 amino acid (Supplementary Fig. S1A). The cDNA of *PtTRII* was 981 bp in length that encoded a 29.1 kDa polypeptide with 261-amino-acid (Supplementary Fig. S1B). BLASTP analysis indicated that both PtTRI and PtTRII belonged to the protein family of NADPH-dependent short-chain dehydrogenases/reductases (SDRs) [20]. At the sequence level, PtTRI and PtTRII showed similarity to other tropine forming reductase and pseudotropine forming reductase (Fig. 2) respectively, such as *H. niger* and *Datura stramonium*. Further, multiple alignments showed that there was a 62.4% similarity between PtTRI and PtTRII at the amino acid sequence level. Both PtTRI and PtTRII had the SDR-specific motifs. In the NADPH-binding site, the conserved TGXXGXG motif was present at the 27–34 position in PtTRI and at the 15–22 position in PtTRII. According to the phylogenetic analysis, tropinone reductases of *Solanaceae* were unequivocally classified into two groups. One group was tropine-forming reductases (TRI), in which the functionally identified TRI proteins from *H.*

niger and *D. stramonium* were included and the other one was pseudotropine-forming reductases (TRII), which included TRII proteins functionally characterized from *H. niger* and *D. stramonium*. Despite the high amino acid sequence similarity between PtTRI and PtTRII, using phylogenetic analysis they were grouped into the TRI and TRII clade (Fig. 3A).

3.2. Tissue expression profiling

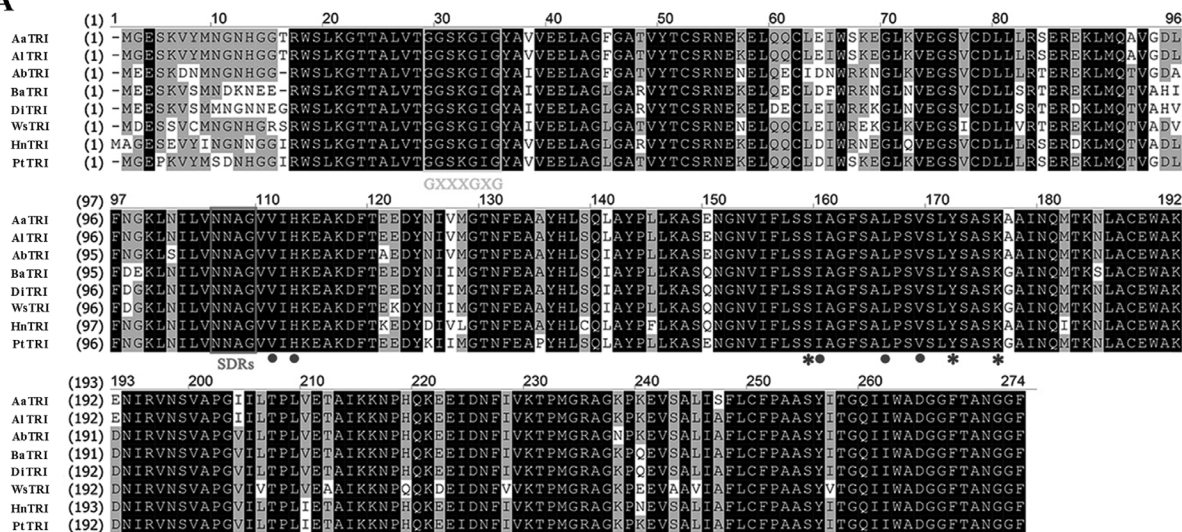
The expression levels of PtTRI and PtTRII genes in *P. tangutica* root, stem and leaf tissues were analyzed by qPCR using cGMP-dependent protein kinase I (PGK) as the internal reference gene. The relative gene expression of *PtTRI* and *PtTRII* in each tissue was calculated and the method of $2^{-\Delta\Delta CT}$ was used to calculate the relative gene expression levels. Fluorescence quantification showed that *PtTRI* was expressed in all three tissues, but the expression levels in different parts of the plant were different and with a higher level in roots (Fig. 3B). Similarly, *PtTRII* was also expressed in all three tissues, and the expression level showed a downward trend. The expression level in roots was the highest, the expression level in stems was about half of that in roots, and the expression in leaves was very small. As the competitive enzyme against PtTRI, *PtTRII* showed different expression pattern. The expression of *PtTRII* was found in roots, stems, and leaves, but with higher level in stems compared with *PtTRI* (Fig. 3C).

3.3. Protein purification and characterization of its catalytic products

Since the amino acid sequences of PtTRI and PtTRII were very similar, enzymatic assays should be performed to respectively identify their enzymatic functions. The coding region of *PtTRI* and *PtTRII* was subcloned into the protein expression vector pET28a and then overexpressed in *E. coli* Rosetta. Recombinant PtTRI and PtTRII were purified. On SDS-PAGE, the molecular weight of PtTRI was about 31.2 kDa (Fig. 4A), and that of PtTRII was approximately 29.1 kDa (Fig. 4B), consistent with their calculated molecular weight. The purified recombinant proteins were then used for enzymatic assays.

When tropinone was fed to the recombinant PtTRI and PtTRII, PtTRI catalyzed the reduction of tropinone to tropine and PtTRII catalyzed the reduction of tropinone reduction to pseudotropine. The products were confirmed by GC–MS (Fig. 5). The retention time for tropine from PtTRI (6.535 Min) was the same (6.670 Min) as that of tropinone standards (Fig. 5A) and the retention time of the product produced by PtTRII (6.915 Min) was the same (6.940 Min) as that of pseudotropine standards (Fig. 5D). The chemical structures were characterized by MS spectrometric data together with Kovats retention indices (RI), quasi-molecular ions ($[M]^+$), and characteristic ions with relative intensity. According to the mass-to-charge ratio of the figure and the structures identified in the NIST Atomic Structure Databases, we demonstrate that PtTRI can catalyze the formation of tropine from the substrate tropinone and that PtTRII had a reducing activity and can reduce tropinone to pseudotropine.

A



B

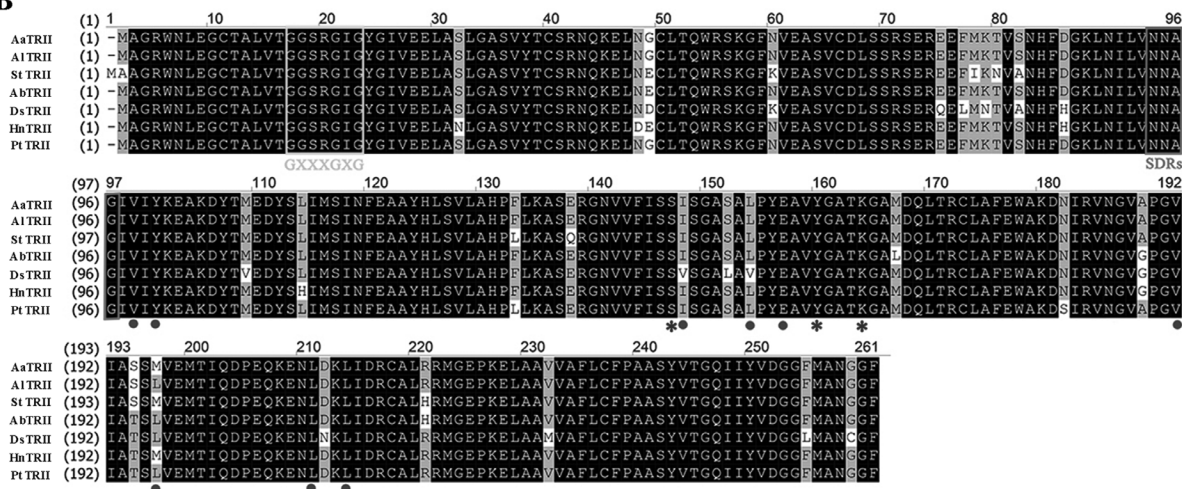
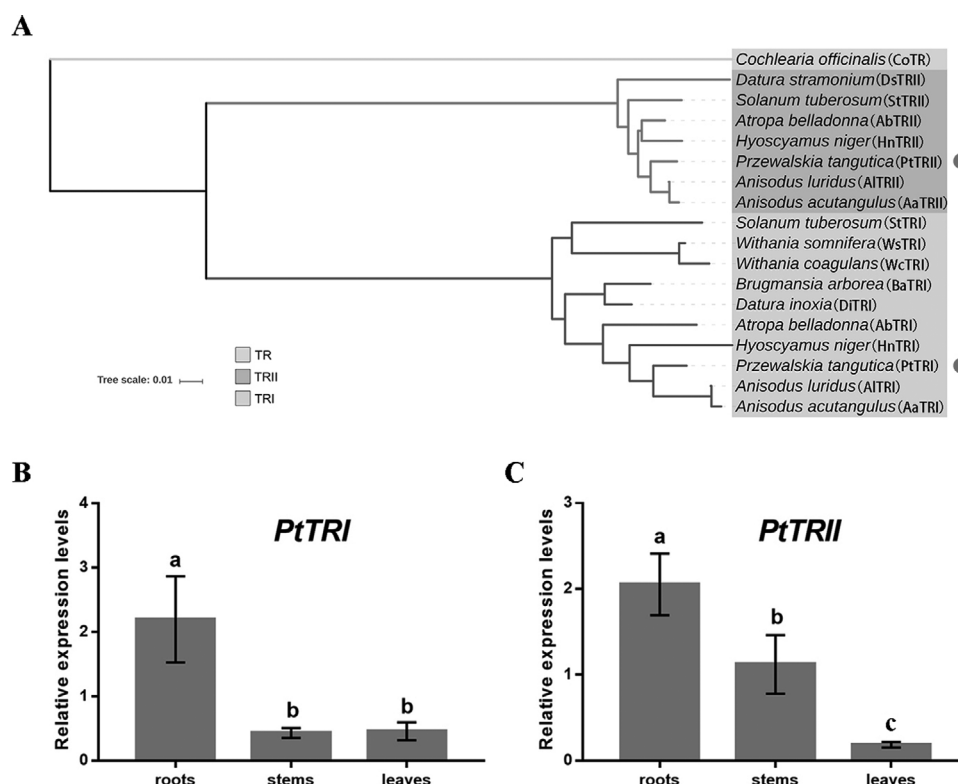


FIG. 2

Protein alignments of TRI and TRII. (A) The amino acid sequences alignment of Solanaceae plant TRI that were functionally identified. AaTRI, *Anisodus acutangulus* TRI; AiTRI, *Anisodus luridus* TRI; AbTRI, *Atropa belladonna* TRI; BaTRI, *Brugmansia arborea* TRI; DiTRI, *Datura innoxia* TRI; WsTRI, *Withania somnifera* TRI; HnTRI, *Hyoscyamus niger* TRI; PtTRI, *Przewalskia tangutica* TRI. (B) The amino acid sequence alignment of Solanaceae plant TRII that were functionally identified. AaTRII, *Anisodus acutangulus* TRII; AiTRII, *Anisodus luridus* TRII; StTRII, *Solanum tuberosum* TRII; AbTRII, *Atropa belladonna* TRII; DsTRII, *Datura stramonium* TRII; HnTRII, *Hyoscyamus niger* TRII; PtTRII, *Przewalskia tangutica* TRII. The identical amino acids are shown in white with black background and the conserved amino acids are shown in black with gray background. Green-decorated residues in green box are involved in NADPH binding. The red box highlight points out residues conserved in SDRs. Amino acids that composing tropinone binding pockets of TRs are also indicated by dark red dots on the bottom. The conserved catalytic residues (S-Y-K) are indicated by blue star on the bottom.


FIG. 3

Phylogenetic tree analysis and tissue profile. (A) Phylogenetic relationship of *PtTRI* and *PtTRII* with other plant TRs. The tree was constructed with maximum likelihood method in MEGA4.1 using sequences from *Solanum tuberosum* TRII (CAC19810.1), *Anisodus luridus* TRII (AGL76990.1), *Anisodus acutangulus* TRII (ACB71203.1), *Atropa belladonna* TRII (AGH24753.1), *Hyoscyamus niger* TRII (AAB09776.1), *Datura stramonium* TRII (AAA33282.1), *Solanum tuberosum* TRI (CAC34420.1), *Anisodus luridus* TRI (AGL76989.1), *Atropa belladonna* TRI (AFP55030.1), *Anisodus acutangulus* TRI (ACB71202.1), *Brugmansia arborea* TRI (AIN39993.1), *Datura innoxia* TRI (AIN39992.1), *Hyoscyamus niger* TRI (BAA13547.1), *Withania somnifera* TRI (AGY46257.1), *Withania coagulans* TRI (AGB56644.1), *Cochlearia officinalis* TRI (CAO02390.1). The numbers on the phylogenetic tree were calculated by 1,000 repeats. (B and C) Expression profiles of *PtTRI* and *PtTRII* in three different tissues. The vertical bars represent mean $\pm$ standard errors (n $\geq$ 3).

3.4. Enzymatic activities of *PtTRI* and *PtTRII* at different pH

In order to determine the optimum pH of *PtTRI* and *PtTRII*, the enzymatic kinetics of *PtTRI* and *PtTRII* on tropinone at different pH conditions was measured. The results showed that *PtTRI* can maintain the reducing activity of tropinone in a wide pH range (pH 5.8–8.4) (Fig. 6A) and the optimum pH point of 6.4. The reduction of tropinone by *PtTRI* had a high catalytic activity in the weakly acidic to weakly alkaline range. It was

found that *PtTRII* can reduce tropinone to pseudotropine with an optimum pH of 6.8 (Fig. 6B).

3.5. Enzymatic kinetics of *PtTRI* and *PtTRII*

The catalytic activities of the two propanone reductases were compared at the optimal pH of *PtTRI* (pH = 6.4) since there is only a slight difference in the reaction rate of *PtTRII* between pH 6.4 and 6.8. Tropinone was selected as the substrate for the reduction reaction and NADPH as the coenzyme. The reaction condition was a potassium phosphate buffer having the same pH of 6.4. According to the Michaelis-Menten curves of *PtTRI* and *PtTRII* (Figs. 6C and 6D). The K_m and K_{cat}/K_m of *PtTRI* for tropinone were 0.52 ± 0.00 mM and $42.26 \text{ Sec}^{-1} \text{ mM}^{-1}$, respectively, and 0.087 ± 0.02 mM and $116.90 \text{ Sec}^{-1} \text{ mM}^{-1}$ for *PtTRII* (Table 1). The affinity and catalytic efficiency of *PtTRII* for tropinone were much higher than for *PtTRI*, (5.9-fold and 2.76-fold higher, respectively). Comparison of the substrate affinity and catalytic efficiency of TRI in the TAs-producing plants demonstrates that recombinant *PtTRI* has a higher catalytic efficiency for tropinone than TRI (Table 2).

4. Discussion

In this study, tropinone reductase I (*PtTRI*) and tropinone reductase II (*PtTRII*) were cloned from *P. tangutica* using RACE technology. In all of the Solanaceae species, the amino acids of TRI and TRII are almost 64% similar and are thought to come from the same ancestor [14, 24]. The similarity in amino acid sequences among *P. tangutica*, *Anisodus acutangulus*,

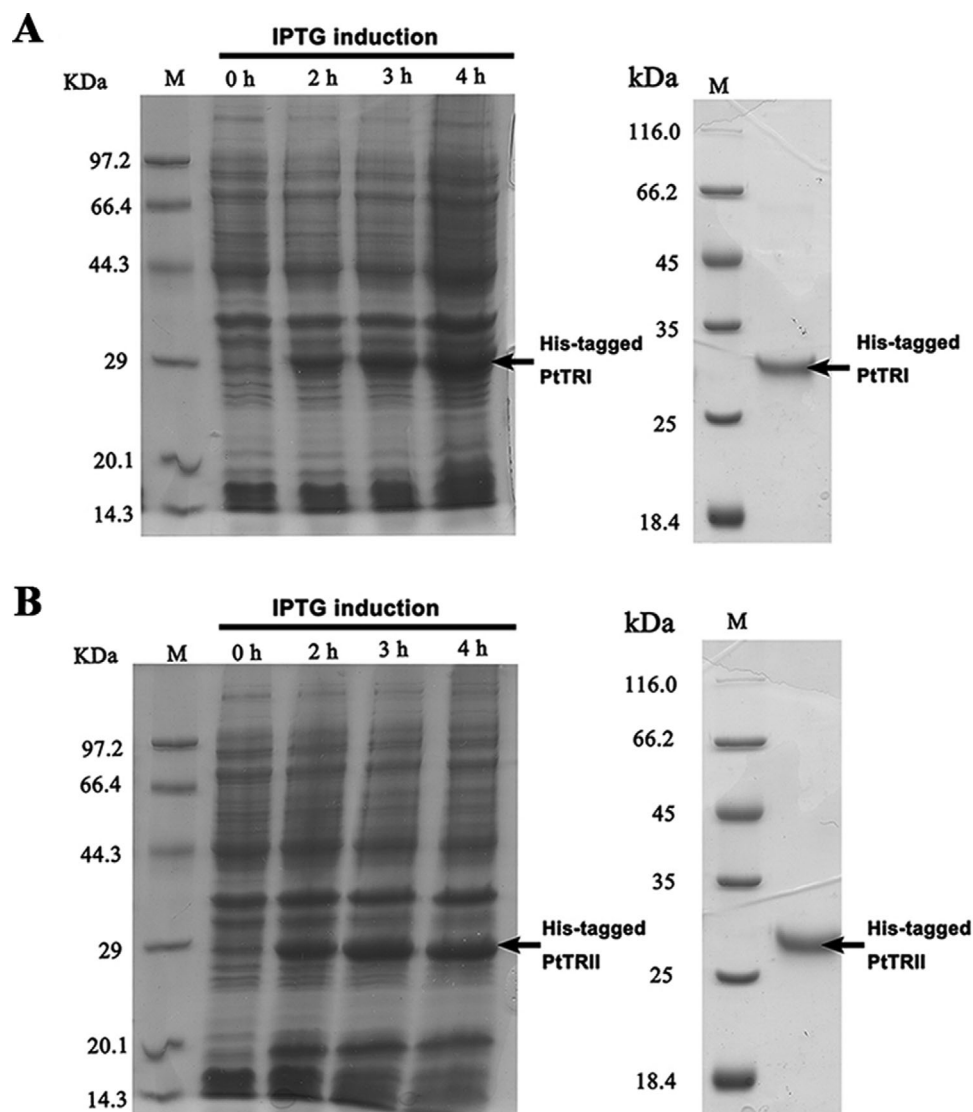


FIG. 4

SDS-PAGE analysis of affinity-purified His-tagged recombinant PtTRI and His-tagged PtTRII. (A) Expression of His-tagged PtTRI from engineered *E. coli*; purifying the His-tagged recombinant PtTRI from engineered *E. coli*. (B) Expression of His-tagged PtTRII from engineered *E. coli*; purifying the His-tagged recombinant PtTRII from engineered *E. coli*.

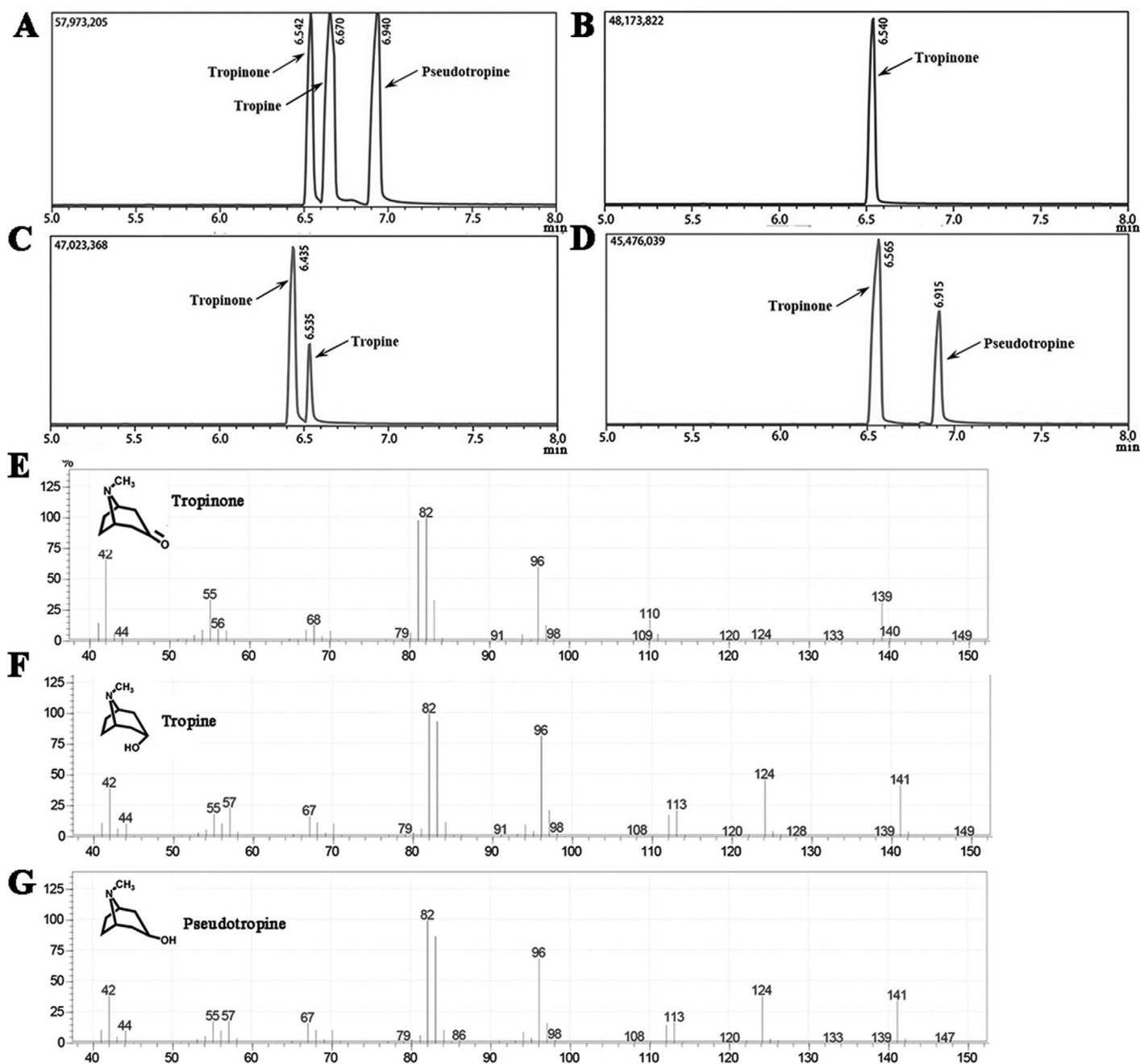
and *Anisodus luridus* is probably due to their geographical proximity. They are all grown at high-altitude in Qinghai Tibet and are popular medicinal plants [4].

Both PtTRI and PtTRII have a conserved NADPH-binding site with a typical sequence characterized by the GXXXGXG motif [25]. There are also two conserved domains in the amino acid sequence: the NNAG domain that is unique to the short-chain dehydrogenase family and the S-Y-K structure which is unique to TRs [26]. The above three conserved structures are essential for the catalytic activity of tropinone reductase. Based on amino acid sequence similarity, it was difficult to identify

the tropine-forming reductase and pseudotropine-forming reductase structures. However, phylogenetic tree construction analysis clearly identified each structure.

Gene expression analysis by qPCR revealed that the *Pt-TRI* transcript was expressed in all three plant organs of *P. tangutica*, with the highest expression level in roots and relatively lower abundance on stems and leaves, which has a similar expression level than other TAs producing plants related genes *PMT* and *H6H* [27], whereas *PtTRII* has a higher expression level in roots and stems. According to the tissue profiles of *PtTRI* and *PtTRII*, it could be deduced that TRI of *P. tangutica* might participate in tropine-derived alkaloid biosynthesis. The *PtTRI* expression level suggests that the biosynthesis of TAs occurred not only in the roots but also in the aerial parts of *P. tangutica*.

Currently, in most Solanaceae species, TRII has a higher affinity for tropinone than TRI as for example in *H. niger* and *D. stramonium* [2, 19]. This study is in accordance with these results, PtTRII catalyzed the reduction of tropinone to form


FIG. 5

Confirmation of tropine by the GC-MS analysis. (A) The GC trace of the standards of tropinone, tropine, and pseudotropine with retention time 6.542, 6.670, and 6.940 Min, respectively. (B) The GC tract of authentic tropinone with retention time 6.540 Min. (C) The GC trace of the reaction buffer showed the substrate tropinone (6.435 Min) and the product tropine (6.535 Min) given by PtTRI. (D) The GC trace of the reaction buffer showed the substrate tropinone (6.565 Min) and the product pseudotropine (6.915 Min) given by PtTRII. (E) The MS data of tropinone. (F) The MS data of tropine. (G) The MS data of pseudotropine.

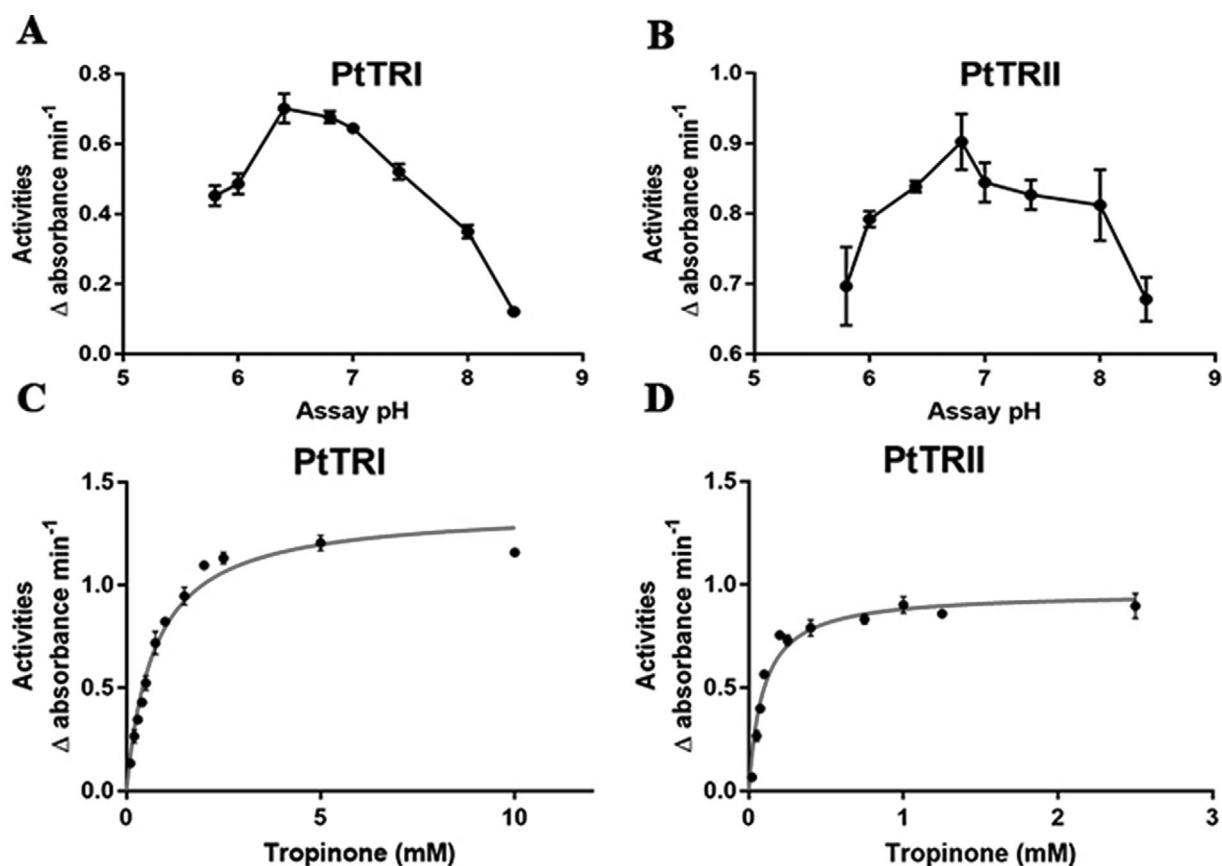


FIG. 6

The optimum pH points and enzymatic activities. (A) Reduction reaction activities of PtTRI at different pH conditions. Each point is the mean of triplicate assays. (B) Reduction reaction activities of PtTRII at different pH conditions. Each point is the mean of triplicate assays. (C) The Michaelis-Menten curves of PtTRI catalyzing tropinone to produce tropine (D) The Michaelis-Menten curves of PtTRII catalyzing the reduction of tropinone to pseudotropine.

TABLE 1

Enzyme kinetics of PtTRI and PtTRII

Tropinone reductase	pH assay	K_m (mM)	V_{max} (nKat μg^{-1})	K_{cat} (Sec^{-1})	K_{cat}/K_m ($\text{Sec}^{-1} \text{mM}^{-1}$)
PtTRI	6.4	0.52 ± 0.10	677.62 ± 21.10	22.36 ± 0.69	42.26
PtTRII	6.4	0.087 ± 0.02	310.80 ± 32.97	10.26 ± 1.08	116.90

pseudotropine with higher affinity to tropinone and higher catalytic efficiency than PtTRI, which catalyzed the reduction of tropinone to tropine. Suppression of SlTRII has been used as an alternative method to enhance the hyoscyamine production in Solanaceae species. It has been demonstrated that

overexpression of TRI can increase the content of hyoscyamine and scopolamine [28, 29], suggesting that PtTRI is a potential candidate for modifying the TAs biosynthetic pathway on TRI to obtain high yielding scopolamine medicinal plants.

TABLE 2

Comparison between various tropinone reductases belonging to the Solanaceae family

Enzyme	Species	K_m (mM)	Reference
PtTRI	<i>Przewalskia tangutica</i>	0.52	In this study
DsTRI	<i>Datura stramonium</i>	1.30	Portsteffen et al. [19]
BaTRI	<i>Brugmansia arborea</i>	2.62	Qiang et al. [21]
DsTRI	<i>Datura stramonium</i>	4.18	Qiang et al. [21]
HnTRI	<i>Hyoscyamus niger</i>	1.01	Hashimoto et al. [2]
PtTRII	<i>Przewalskia tangutica</i>	0.087	In this study
HnTRII	<i>Hyoscyamus niger</i>	0.034	Hashimoto et al. [2]
AbTRII	<i>Atropa belladonna</i>	0.090	Dräger and Schaal [22]
StTRII	<i>Solanum tuberosum</i>	0.033	Kaiser et al. [23]
DsTRII	<i>Datura stramonium</i>	0.033	Portsteffen et al. [19]

5. Acknowledgements

This work was financially supported by the NSFC projects (31370333 and 31770335), the CSTC project (cstc2017jcyjAX0208), and the YNTC Foundation (2016YN22).

6. Author Contributions

N.W. and Z.L. conceived and designed the study. N.W., D.J., and M.X. performed gene cloning, expression analysis, and biochemical assays. M.C. and X.L. performed bioinformatics analysis. N.W. and X.L. prepared the manuscript. All authors have read and approved the manuscript.

The authors declare no conflict of interest.

7. References

- [1] Xiao, P. G. and He, L. Y. (1982) *Planta Med.* 45, 112–115.
- [2] Hashimoto, T., Nakajima, K., Ongena, G., and Yamada, Y. (1992) *Plant Physiol.* 100, 836–845.
- [3] akabová, S., Vincze, L., Farkas, Á., Kilár, F., Boros, B., and Felinger, A. (2012) *J. Chromatogr. A* 1232, 295–301.
- [4] Zhao, K. H., Zeng, J. L., Zhao, T. F., Zhang, H. X., Qiu, F., Yang, C. X., Zeng, L. J., Liu, X. Q., Chen, M., Lan, X. Z., and Liao, Z. H. (2017) *Front. Plant Sci.* 8, 1745.
- [5] Lan, X. Z. and Quan, H. (2010) *J. Med. Plants Res.* 4, 1477–1481.
- [6] Dräger, B. (2006). *Phytochemistry* 67, 327–337.
- [7] Nakajima, K. and Hashimoto, T. (1999) *Plant Cell Physiol.* 40, 1099–1107.
- [8] Nakajima, K., Oshita, Y., Yamada, Y., and Hashimoto, T. (1999) *Biosci. Biotechnol. Biochem.* 63, 1819–1822.
- [9] Wang, X. R., Chen, M., Yang, C. X., Liu, X. Q., Zhang, L., Lan, X. Z., Tang, K. X., and Liao, Z. H. (2011) *Physiol. Plantarum.* 143, 309–315.
- [10] Kang, Y. M., Park, D. J., Min, J. Y., Song, H. J., Jeong, M. J., Kim, Y. D., Kang, S. M., Karigar, C. S., and Choi, M. S. (2011) *In Vitro Cell. Dev. Biol. Plant* 47, 516–524.
- [11] Shi, P., Fu, X. Q., Liu, F. M., Shen, Q., Jiang, W. M., Li, L., Sun, X. F., and Tang, K. X. (2017) *Plant Cell Tiss.* 129, 251–259.
- [12] Tang, K. X., Shen, Q., Yan, T., and Fu, X. Q. (2014) *Plant Cell Rep.* 33, 605–615.
- [13] Lv, Z., Zhang, F. Y., Pan, Q. F., Fu, X. Q., Jiang, W. M., Shen, Q., Yan, T. X., Shi, P., Lu, X., Sun, X. F., and Tang, K. X. (2016) *Plant Cell Physiol.* 57, 588–602.
- [14] Kushwaha, A. K., Sangwan, N. S., Trivedi, P. K., Negi, A. S., Misra, L., and Sangwan, R. S. (2013) *PloS One* 8, e74777.
- [15] Dräger, B., Hashimoto, T., and Yamada, Y. (1988) *Agric. Biol. Chem.* 52, 2663–2667.
- [16] Nakajima, K., Hashimoto, T., and Yamada, Y. (1993) *Plant Physiol.* 103, 1465–1466.
- [17] Richter, U., Rothe, G., Fabian, A. K., Rahfeld, B., and Dräger, B. (2005) *J. Exp. Bot.* 56, 645–652.
- [18] Livak, K. J. and Schmittgen, T. D. (2001). *Methods* 25, 402–408.
- [19] Portsteffen, A., Dräger, B., and Nahrstedt, A. (1994) *Phytochemistry* 37, 391–400.
- [20] Persson, B., Kallberg, Y., Oppermann, U., and Jörnvall, H. (2003) *Chem-Biol. Interact.* 18, 271–278.
- [21] Qiang, W., Xia, K., Zhang, Q. Z., Zeng, J. L., Huang, Y. S., Yang, C. X., Chen, M., Liu, X. Q., Lan, X. Z., and Liao, Z. H. (2016) *Phytochemistry* 127, 12–22.
- [22] Dräger, B., and Schaal, A. (1994) *Phytochemistry* 35, 1441–1447.
- [23] Kaiser, H., Richter, U., Keiner, R., Brabant, A., Hause, B., and Dräger, B. (2006) *Planta* 225, 127–137.
- [24] Nakajima, K., Hashimoto, T., and Yamada, Y. (1993) *Proc. Natl. Acad. Sci. USA* 90, 9591–9595.
- [25] Chen, W., Cheng, X. F., Zhou, Z. H., Liu, J. J., and Wang, H. Z. (2012) *Mol. Biol. Rep.* 40, 1145–1154.
- [26] Kai, G. Y., Li, L., Jiang, Y. X., Yan, X. M., Zhang, Y., Lu, X., Liao, P., and Chen, J. B. (2009) *Biotechnol. Appl. Biochem.* 54, 177–186.
- [27] Cheng, X. F., Chen, W., Zhou, Z. H., Liu, J. J., and Wang, H. Z. (2013) *J. Plant Physiol.* 170, 958–964.
- [28] Kai, G. Y., Zhang, A., Guo, Y. Y., Li, L., Cui, L. J., Luo, X. Q., Liu, C., and Xiao, J. B. (2012) *Mol. Biosyst.* 8, 2883–2890.
- [29] Kai, G. Y., Yang, S., Luo, X. Q., Zhou, W. T., Fu, X. Q., Zhang, A., Zhang, Y. Z., and Xiao, J. B. (2011) *BMC Biotechnol.* 11, 43–43.